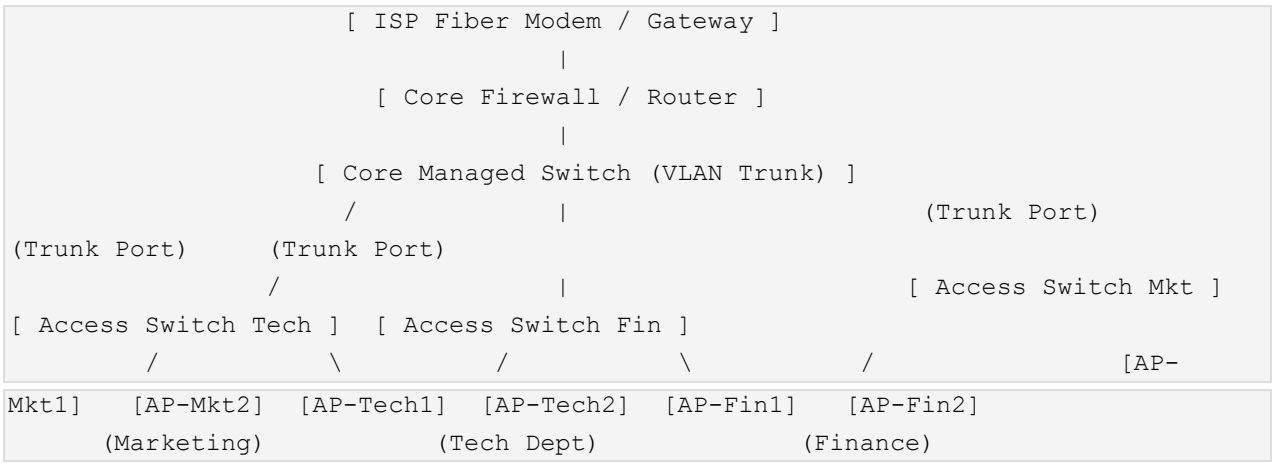
DETAILED DESIGN & HARDWARE SELECTION

1. Draw a detailed network topology diagram for a small co-working space with 3 departments (e.g., Marketing, Tech, Finance), showing routers, switches, wireless access points, and user devices using any diagram tool like draw.io or Lucidchart.

Answer & Topological Layout:



Department	VLAN ID	Subnet / CIDR	Default Gateway	Sample Device IPs
Marketing	VLAN 10	192.168.10.0/24	192.168.10.1	.10 (Printer), .11 (Laptop 1), .12 (Laptop 2), .13 (Desktop), .14 (Tablet)
Tech	VLAN 20	192.168.20.0/24	192.168.20.1	.20 (Server), .21 (Dev PC 1), .22 (Dev PC 2), .23 (Test Rig), .24 (MacBook)
Finance	VLAN 30	192.168.30.0/24	192.168.30.1	.30 (NAS), .31 (CFO PC), .32 (Accountant 1), .33 (Accountant 2), .34 (Printer)

Description: The core firewall connects to a central managed switch running 802.1Q VLAN trunking. Each department (Marketing, Tech, Finance) features dedicated access switches and Wi-Fi access points to isolate traffic.

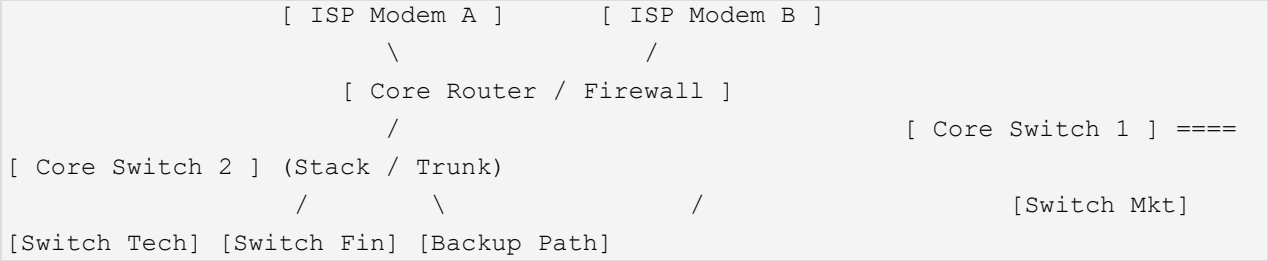
2. Create an IP addressing spreadsheet for the above co-working space, assigning uniquesubnets and VLAN IDs to each department, and listing at least 5 device IPs per department.

Answer & IP Addressing Table:

3. Design a basic redundancy plan for the network that ensures internet connectivity isavailable even if one main switch fails. Briefly describe your approach and update your topology diagram to include redundancy.

Answer & Redundancy Design:

Approach: To eliminate single points of failure, we deploy dual core switches configured with Spanning Tree Protocol (STP / RSTP) and Link Aggregation (EtherChannel) for uplinks. If Access Switch A fails, redundant backup trunk lines automatically reroute traffic through secondary pathways.



4. Select suitable network hardware (router, switches, access points) for the co-workingspace scenario and create a Bill of Materials (BOM) table with device model, quantity, and estimated price (use online prices from Indian retailers like Amazon or Flipkart).

Answer & Bill of Materials (BOM):

Category	Model	Qty	Unit Price (INR)	Total Price (INR)
Firewall / Router	Fortinet FortiGate 40F	1	₹45,000	₹45,000
Core Managed Switch	TP-Link JetStream TL-SG3428	2	₹12,000	₹24,000
Access Points	TP-Link EAP650 (Wi-Fi 6 AP)	3	₹8,500	₹25,500
Total Estimated Cost				**₹94,500**

5. Draft a short security policy for the co-working space network covering at least: WiFipassword rules, VLAN access restrictions, guest network setup, and basic firewall guidelines.

Answer & Security Policy Draft:

- **1. Wi-Fi Password Rules:** Staff Wi-Fi requires WPA3-Enterprise authentication with individual user credentials. Passphrases must be at least 14 characters long and rotated every 90 days.
- **2. VLAN Access Restrictions:** Inter-VLAN routing is strictly blocked by default on the firewall. Marketing, Tech, and Finance can only communicate with the internet and authorized internal shared printers via explicit firewall rules.
- **3. Guest Network Setup:** Public guest Wi-Fi operates on an isolated VLAN (VLAN 99) with client isolation enabled, preventing guests from communicating with each other or accessing internal corporate subnets.

- **4. Basic Firewall Guidelines:** Default-deny policy for all inbound external traffic. Intrusion Prevention Systems (IPS) and web filtering must remain active across all corporate gateways to mitigate malware and unauthorized data exfiltration.